

Fundamentals of Data Science — 12-Week Course Plan

Bachelor's Term 3 • 2h lecture + 2h lab weekly

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1. What you have

Everything for this course is in one folder. Nothing else is required, and nothing needs to be purchased.

Give to students on day one

File	Pages	What it is
FDS_Volume_1_Main_Text.docx	127	The textbook: 15 chapters, 45 worked examples, 28 figures, 60 exercises
FDS_Volume_2_Reference.docx	48	Model answers, code cookbook, formula sheet, decision tables, glossary, dataset documentation
Lab_Manual.docx	32	Lab 0 preparation plus twelve two-hour labs
data/	—	Seven CSV files, synthetic, CCO
data/ DATA_DICTIONARY.md	—	Column-level documentation

Students put the data folder in Google Drive as MyDrive/fds/data/. Everything else runs in Google Colab. There is nothing to install.

Keep to yourself

File	Why
Course_Handbook.docx	Contains the four planted structures in plain language
make_dataset.py	Its docstring reveals the same. Withhold until after Week 6
test_book_code.py	Verifies ~90 numbers in the book against the data. Run it if you change anything
figures/	All 28 book figures, if you want them for

File	Why
	slides

The dataset

File	Rows	One row is
students.csv	1,200	one student
enrolments.csv	4,665	one student-course registration
activity.csv	16,165	one student-week of platform activity
outcomes.csv	1,200	one student, weeks 1–4 features plus the withdrawal target
feedback.csv	732	one free-text course comment
platform_daily.csv	364	one day of platform sessions
students_dirty.csv	1,205	students.csv with realistic damage, for the cleaning week

Synthetic, fixed seed, CCO, no real student record. The same 1,200 students appear in Week 1 and Week 12, so effort spent understanding them early keeps paying.

2. Learning outcomes

	Outcome	Weeks	Assessed by
CL01	Explain fundamental concepts: collection, preprocessing, EDA, basic statistics	1–8	Labs 1–8, Data Quality Report, EDA Brief
CL02	Analyse numerical, categorical, time-series and text data; clean and transform appropriately	1–5, 12	Labs 1–5, 12, Data Quality Report
CL03	Apply supervised and unsupervised techniques for insight and decision	9–11	Labs 9–11, Modelling Assignment

	Outcome	Weeks	Assessed by
CLO4	support Communicate through visualization, dashboards and reports, with ethical and privacy awareness	6, 10, 12	EDA Brief, Modelling Assignment, Capstone, Presentation

Ethics is not a week of its own. It appears in Week 1 (framing and consent), Week 3 (what cleaning deletes), Week 6 (honest charts), Week 10 (thresholds and disparate impact), Week 11 (naming people), and Week 12 (privacy and contestability). A single ethics lecture at the end teaches students that ethics is a compliance step.

3. The twelve weeks

Wk	Lecture (2h)	Chapters	Lab (2h)	CLO
0	—	—	Lab 0: preparation, done independently	—
1	Framing questions; the claim ladder; lifecycle; sources, sampling frames, proxies	1, 2	Lab 1 — Rows, joins, and what is missing	1, 2
2	Centre, spread, shape, standardisation	3	Lab 2 — Summaries by hand, then in code	1
3	Quality dimensions; MCAR/MAR/MNAR; duplicates; outliers	4	Lab 3 — Clean the registry export	1, 2
4	Scaling, encoding, date features;	5	Lab 4 — Pipelines and leakage	2

Wk	Lecture (2h)	Chapters	Lab (2h)	CLO
	leakage; pipelines			
5	Structured EDA; confounding; Simpson's paradox	6	Lab 5 — Exploratory analysis that survives scrutiny	1, 2
6	Visual encoding; baselines; colour and accessibility; message titles	7	Lab 6 — Chart critique and redesign	4
7	Probability; conditional direction; Bayes with natural frequencies	8	Lab 7 — Probability and simulation	1
8	Sampling variability; confidence intervals; tests; effect size; A/B design	9	Lab 8 — Intervals, resampling, and power	1
9	Baselines; linear regression; metrics; residuals; cross-validation	10	Lab 9 — The regression workflow	3
10	Logistic regression and log-odds; k-NN and trees; confusion matrix; thresholds; ROC vs PR; fairness	11, 12	Lab 10 — Classification, end to end	3, 4
11	Distance; k-means; choosing k; PCA; naming	13	Lab 11 — Clustering with honest uncertainty	3

Wk	Lecture (2h)	Chapters	Lab (2h)	CLO
12	responsibly Time series and text; dashboards; privacy; contestability	14, 15	Lab 12 — Time series, text, and a dashboard	2, 4

What was compressed to fit 12 weeks

The book is written for 14 weeks. Two merges get it to 12.

Week 1 covers Chapters 1 and 2. Both are conceptual with a light technical load, and the first week always loses time to setup anyway. Lab 0 absorbs that setup so the lab itself stays productive.

Week 10 covers Chapters 11 and 12 — building a classifier and evaluating one. This is the heaviest week in the course. Two things make it fit:

- **Drop calibration to a reading.** Section 12.6 is self-contained. Set it as reading and assess it only in the capstone.
- **Protect Task 4 of Lab 10.** The fairness audit is where CLO3 and CLO4 meet, and it is the first thing that gets cut when time runs short. Cut Task 2's hand calculation instead if you must — though that one is the answer to “why was I flagged”, so cut it reluctantly.

If you gain a week, split Week 10 back into two. That is the single best use of extra time in this syllabus.

If you lose a week, merge Weeks 7 and 8 into one statistics week and set the permutation test as homework. Do **not** cut Week 4. Leakage is the idea students most reliably get wrong in the capstone if it was rushed.

4. Assessment

Component	Weight	CLO	Due
Weekly lab notebooks (best 9 of 12)	20%	1, 2, 3	Weekly
Data Quality Report	10%	1, 2	Week 4
EDA and Visualization Brief	15%	1, 2, 4	Week 7

Component	Weight	CLO	Due
Modelling Assignment	20%	3	Week 11
Capstone project and dashboard	25%	1–4	Week 12
Capstone presentation	10%	4	Week 12

Labs are pass / revise / fail, not finely marked. Turnaround speed matters more than precision, and best-9-of-12 absorbs illness without an administrative process.

Every submission from Week 4 must run top to bottom from a restarted notebook. Say this in Week 1; enforce it from Week 4.

The capstone uses a student-chosen public dataset, not the course dataset. The course data are the teaching instrument; the capstone tests whether the habits transfer.

Rubrics

Full rubrics are in the instructor handbook. The two lines that matter most:

- **The threshold justification in the Modelling Assignment carries 25 of its 100 marks.** It is the part students most want to skip and the part that most determines whether their model would harm anyone.
- **The limitations section in the Capstone carries 20 of its 100 marks.** If limitations are not marked, students correctly infer they are decorative.

5. Before Week 1

1. **Run test_book_code.py.** It must end with “All checks passed.” This verifies about 90 numbers quoted in the book against the data.
2. **Open Colab on an actual lab machine** — not your laptop. Some campus networks block Google services, and that is the only failure that breaks everything.
3. **Put the data in your own Drive** and run the Lab 0 setup cell. You should get (1200, 7) (4665, 5) (16165, 6) (1200, 7).
4. **Publish** the two volumes, the Lab Manual, the data, and the data dictionary. Tell students to complete Lab 0 before Week 1.
5. **Prepare Weeks 1–3 fully.** After that, stay three weeks ahead. Preparing a whole term in advance never survives contact with a cohort.

Two rules to state in Week 1

1. **Runtime → Restart and run all** before submitting anything.
2. A number without a sentence interpreting it is not an answer.

Say both in Week 1, repeat weekly, enforce in marking from Week 4. Students who hear this early adapt easily; students who first hear it in Week 9 do not.

6. Where students reliably go wrong

Failure	First appears	Intervention
Counting registrations as students	Week 1	Make them say “one row is one ___” aloud before every count
Merging without checking cardinality	Week 1	Require validate= on every merge from Week 1
Imputing before diagnosing missingness	Week 3	Ban imputation entirely in Lab 3
Fitting a scaler before splitting	Week 4	Require every transformer inside a Pipeline from Week 4
Reporting accuracy on the withdrawal model	Week 10	Have them compute the majority baseline first, every time
Causal language in the capstone	Week 12	Mark the claim rung explicitly in the Week 7 brief so the habit forms early

Week 7 is the split point

Probability and Bayes is where cohorts most often divide, and students who are lost go quiet rather than ask. Two things help: work the screening example with **counts on the board before any notation**, and make the natural-frequency version the assessed form. The formula can come second, or not at all.

Week 8 peer reproduction

In Lab 8, students swap notebooks and try to reproduce each other's Week 5 result. Roughly half of first attempts fail, usually on a Drive path that only exists for the author.

Frame this before they start: **the failure is the lesson and costs no marks**. Otherwise students experience it as an accusation of incompetence rather than the most efficient reproducibility lesson available.

7. Quick reference

12 weeks · 2h lecture + 2h lab · 15 chapters · Lab 0 plus 12 labs

Publish day one: both volumes, Lab Manual, data/, data dictionary.

Never publish: Course_Handbook.docx, and make_dataset.py before Week 6.

Heaviest week: 10. Drop calibration to a reading; protect the fairness audit.

Never cut: Week 4 (leakage).

The one check that prevents most problems: every student has completed Lab 0 by the end of Week 1.